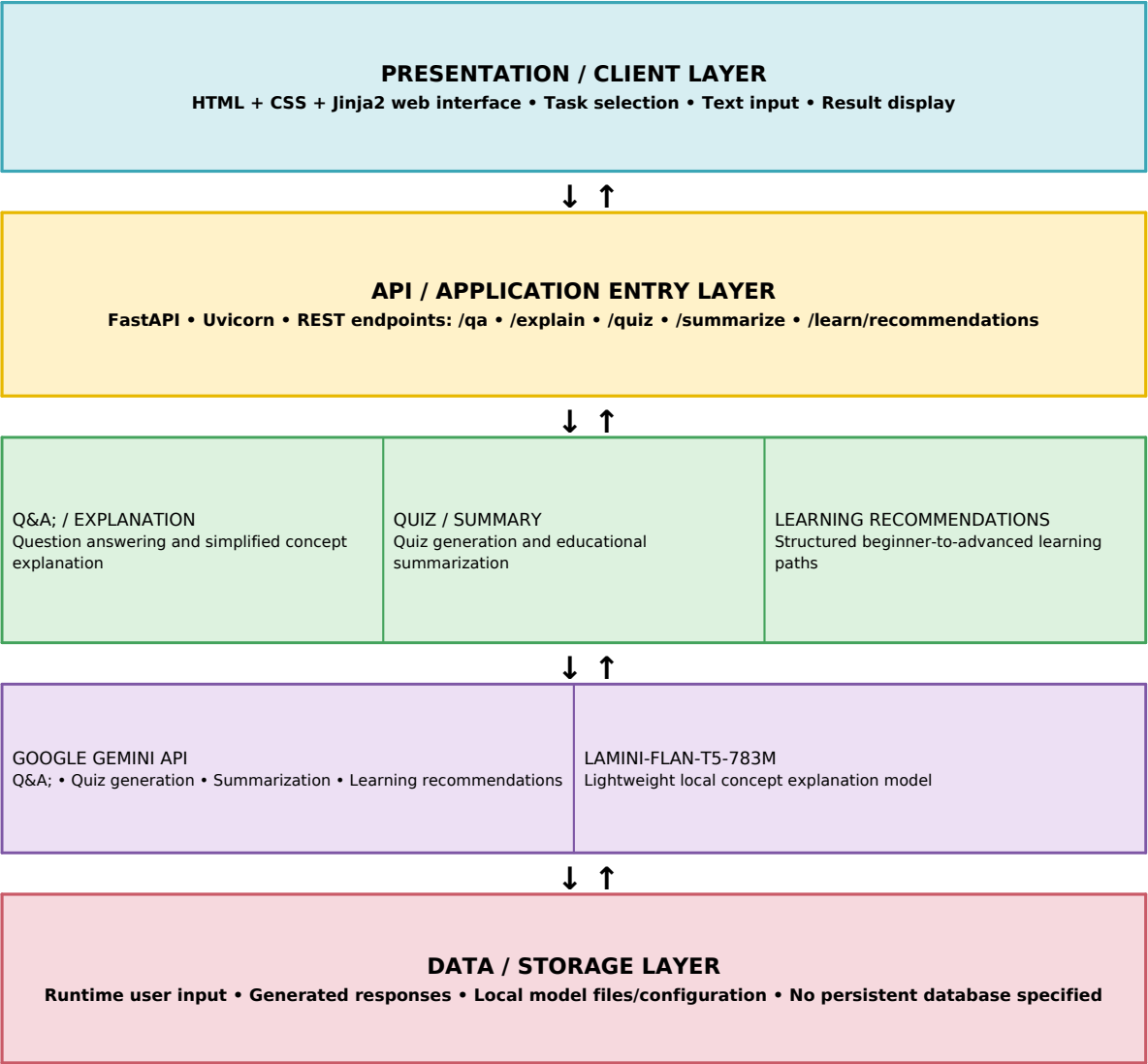


Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-4683
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of the solution's architecture. The structure below maps the Presentation Layer, Application Layer, and Data/Storage Layer to the EduGenie components.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the learner-facing interface for selecting a learning task, entering text and viewing generated results.	HTML, CSS, Jinja2
API / Application Layer	Receives requests, routes them to the appropriate learning module and returns the generated result to the frontend.	Python, FastAPI, Uvicorn
Q&A; Module	Handles learner questions and generates answers using the configured AI service.	Python, Gemini API
Explanation Module	Provides simplified explanations of concepts using a lightweight local language model.	Python, LaMini-Flan-T5-783M
Quiz Module	Generates multiple-choice practice questions from supplied content/topic.	Python, Gemini API
Summarization Module	Converts long educational passages into concise study-oriented summaries.	Python, Gemini API
Learning Recommendation Module	Generates structured learning paths and recommended progression from beginner to advanced concepts.	Python, Gemini API
External AI Integration	Provides cloud generative AI capabilities for the modules that use Gemini.	Google Gemini API / Google AI Studio
Data / Storage	Handles runtime inputs, generated outputs and local model/configuration files. No persistent database is specified.	Local runtime / filesystem

Architecture Flow

Learner → Web Interface → FastAPI Endpoint → Feature Module → Gemini API / LaMini-Flan-T5 → Generated Result → Web Interface.

The architecture is modular: each learning capability is exposed through a separate FastAPI endpoint and can be updated independently.